10A Data Collection, Summary Statistics and Outliers

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. For the 7 ordered data scores $11, 14, 14, 16, 19, 21, 24$, find:

(a) The mode (b) The range (c) The median (d) The mean

Q2. For the 8 unsorted data scores $22, 15, 28, 18, 15, 25, 13, 24$, find:

(a) The mode (b) The range (c) The median (d) The mean

Q3. A frequency table of books read in a month by 20 students shows: $(x = 1, f = 3)$, $(x = 2, f = 8)$, $(x = 3, f = 5)$, $(x = 4, f = 3)$, $(x = 5, f = 1)$. Find:

(a) The modal number of books (b) The range (c) The median number of books (d) The mean number of books ($\bar{x} = \frac{\sum fx}{\sum f}$)

Q4. Classify each data collection scenario:

(a) Testing every battery on a factory assembly line (Census or Sample?)

(b) Polling 500 randomly chosen Victorian voters (Census or Sample?)

(c) Counting cars passing your school gate yourself (Primary or Secondary?)

(d) Using ABS census tables for a geography report (Primary or Secondary?)

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. Seven properties sold in a Ballarat suburb in one week for (in thousands of dollars): $510, 530, 540, 540, 560, 580, 1400$ (where $\$1\,400\,000$ was a heritage acreage estate). Find (a) the median and mean of all 7 prices, and (b) the new median and new mean when the $\$1\,400\,000$ outlier is removed.

Q6. Over his first 5 netball matches, Noah's team scored $42, 38, 45, 51,$ and 39 goals. How many goals must the team score in the 6th match to achieve a mean of 44 goals per match across all 6 matches?

Q7. In a Year 8 science cohort, Class 8A (20 students) has a mean test score of 76, and Class 8B (30 students) has a mean test score of 81. Calculate the combined mean score for all 50 students.

Q8. A stem-and-leaf plot of daily commuter cycling times (in minutes, where $2 \mid 5 = 25$) for 9 trips is: $1 \mid 8, 9$; $2 \mid 1, 2, 4, 5, 7, 8$; $6 \mid 5$. Find the median, mean, and range of the 9 trips, and identify the outlier.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. The mean of 6 numbers is 15. When one number is removed, the mean of the remaining 5 numbers drops to 13. Find the value of the number that was removed.

Q10. Find a set of 4 positive integers written in ascending order $a \le b \le c \le d$ such that the mode is 5, the median is 7, and the mean is 8. Show each step of your deduction.